

影像處理 期中複習

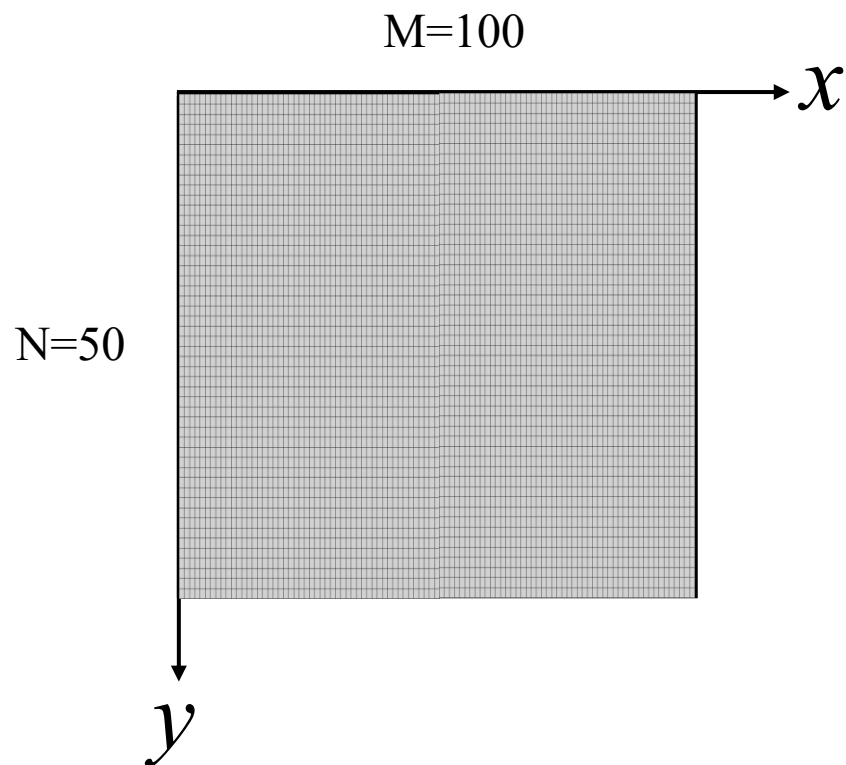
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助教:陳信宏

第2章:索引、雙線性內插、仿射

Chapter 2: Indexing, Bilinear Interpolation, and Affine Transformation

範例一:影像索引2D



Q: ➤ 已知索引值 $\alpha = 120$, 求 (x, y) 座標
➤ Given the index value α , find (x, y) .

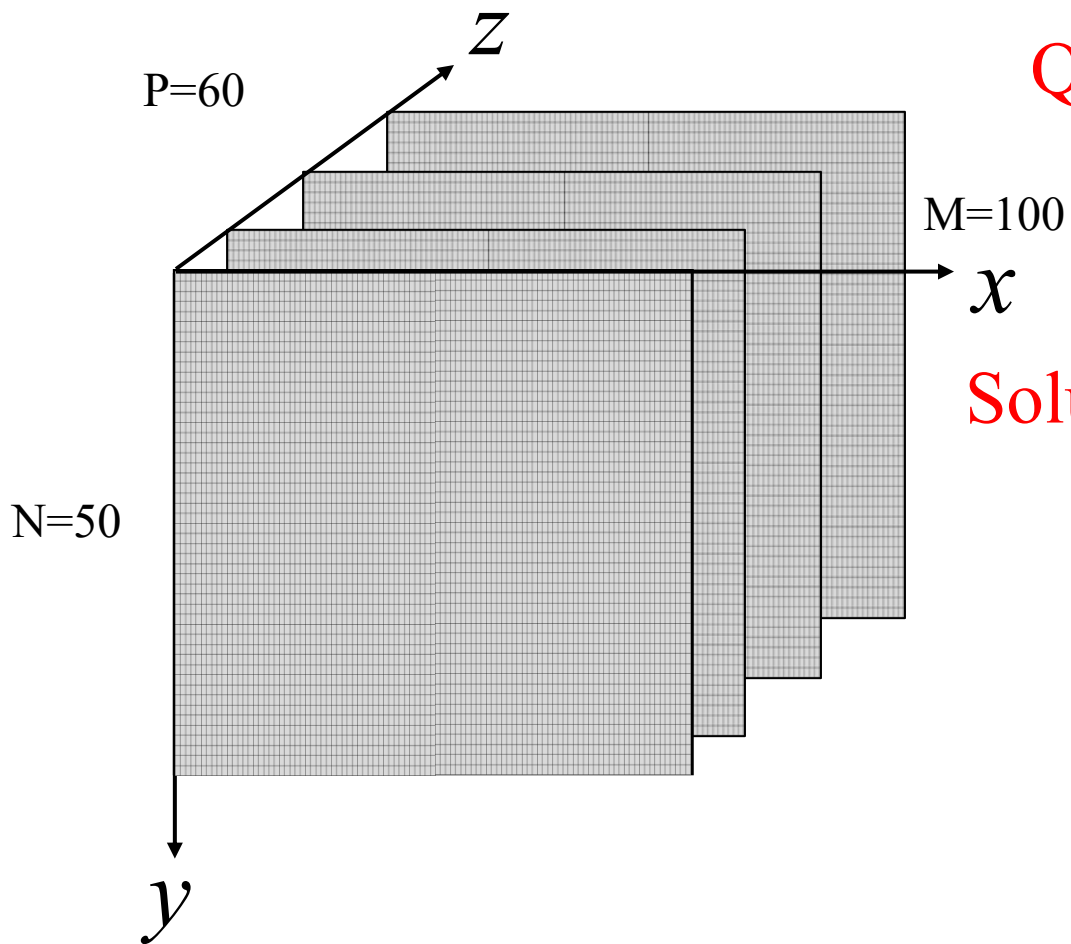
Solution:

$$\begin{aligned} x &= \alpha \bmod M \\ y &= \alpha // M \text{ (取商數)} \end{aligned}$$

$$\begin{aligned} x &= 120 \bmod 100 = 20 \\ y &= 120 // 100 = 1 \end{aligned}$$

Ans: $(x, y) = (20, 1)$ #

範例二:影像索引3D



Q: ➤ 已知索引值 $S = 5090$, 求 (x, y, z) 座標
➤ Given the index value $S = 5090$, find (x, y, z) .

Solution:

$$\begin{aligned} x &= S \bmod MN \bmod M \\ y &= S \bmod MN // M \text{ (取商數)} \\ z &= S // MN \text{ (取商數)} \end{aligned}$$

$$\begin{aligned} x &= 5090 \bmod (100 * 50) \bmod 100 = 90 \\ y &= 5090 \bmod (100 * 50) // 100 = 0 \\ z &= 5090 // (100 * 50) = 1 \end{aligned}$$

Ans: $(x, y, z) = (90, 0, 1)$ #

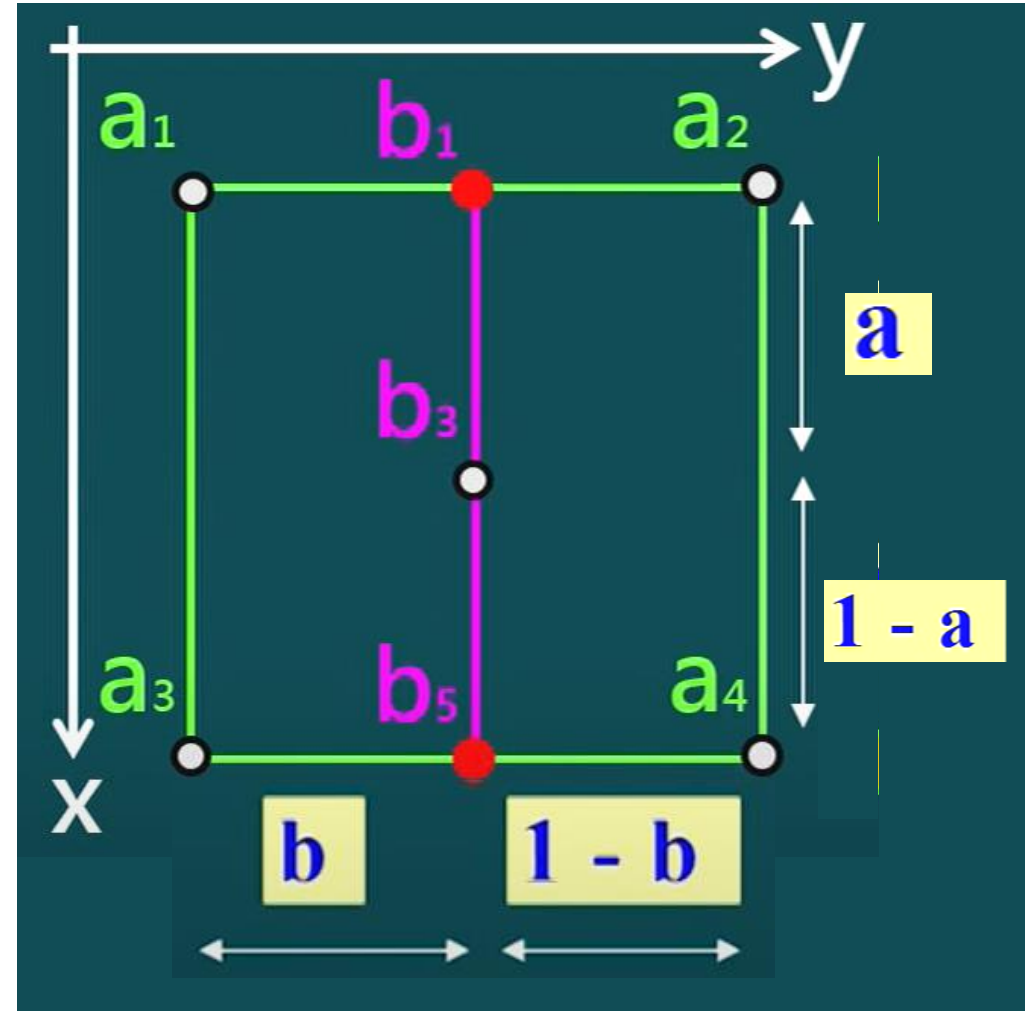
★ 2-D 雙線性內插法(Bilinear Interpolation)

1-D線性內插法

$$f(n + a) = (1 - a) \times f(n) + a \times f(n + 1)$$

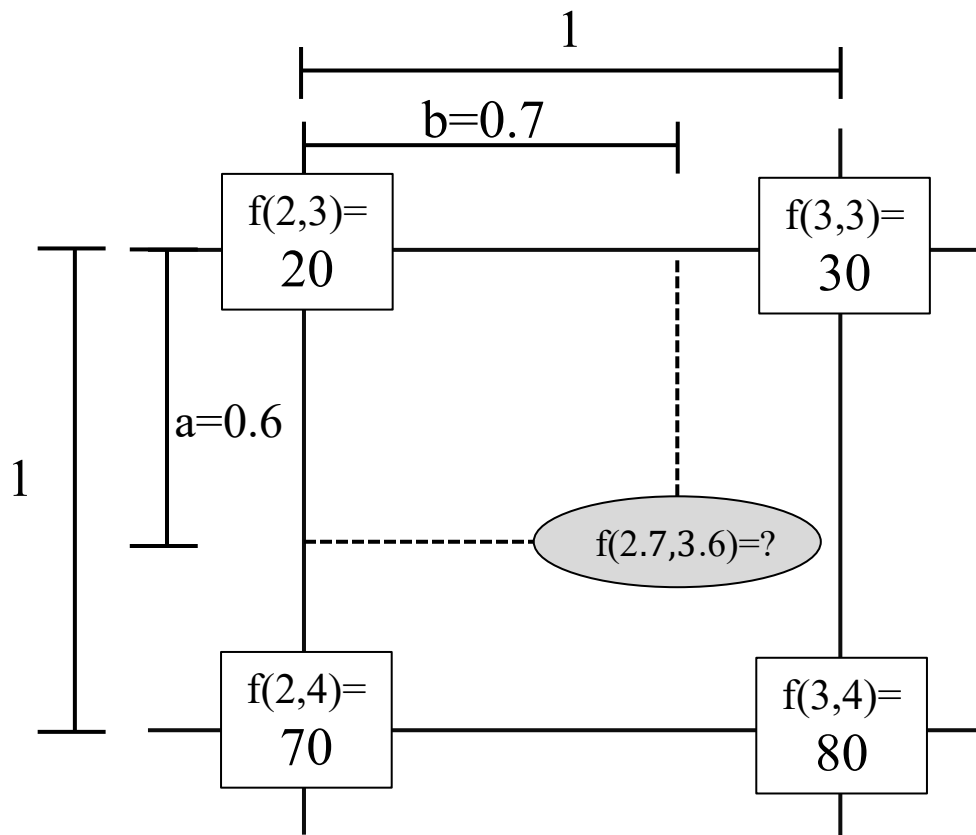
2-D線性內插法

$$b_3 = (1 - a) \times [(1 - b) \times a_1 + b \times a_2] \\ + a \times [(1 - b) \times a_3 + b \times a_4]$$



範例三:雙線性內插(Bilinear interpolation)

- Q:** ➤ 利用雙線性內插法，計算 $f(2.7, 3.6)$ 的影像強度。
- Use bilinear interpolation to compute $f(2.7, 3.6)$ given the four corner values.



Sol:

$$b_1 = (1 - b) \times a_1 + b \times a_2$$

$$b_5 = (1 - b) \times a_3 + b \times a_4$$

$$b_3 = (1 - a) \times b_1 + a \times b_5$$

$$0.3 \times 20 + 0.7 \times 30 = 27$$

$$0.3 \times 70 + 0.7 \times 80 = 77$$

$$0.4 \times 27 + 0.6 \times 77 = 57$$

Ans: $f(2.7, 3.6) = 57$ #



2-D 雙線性內插法 整理

Step1:縮放比例 (Scaling Ratio)

- 原始圖像尺寸:寬度 w_1 , 高度 h_1
- 目標圖像尺寸:寬度 w_2 , 高度 h_2

$$S_x = \frac{w_1 - 1}{w_2 - 1} \quad S_y = \frac{h_1 - 1}{h_2 - 1}$$

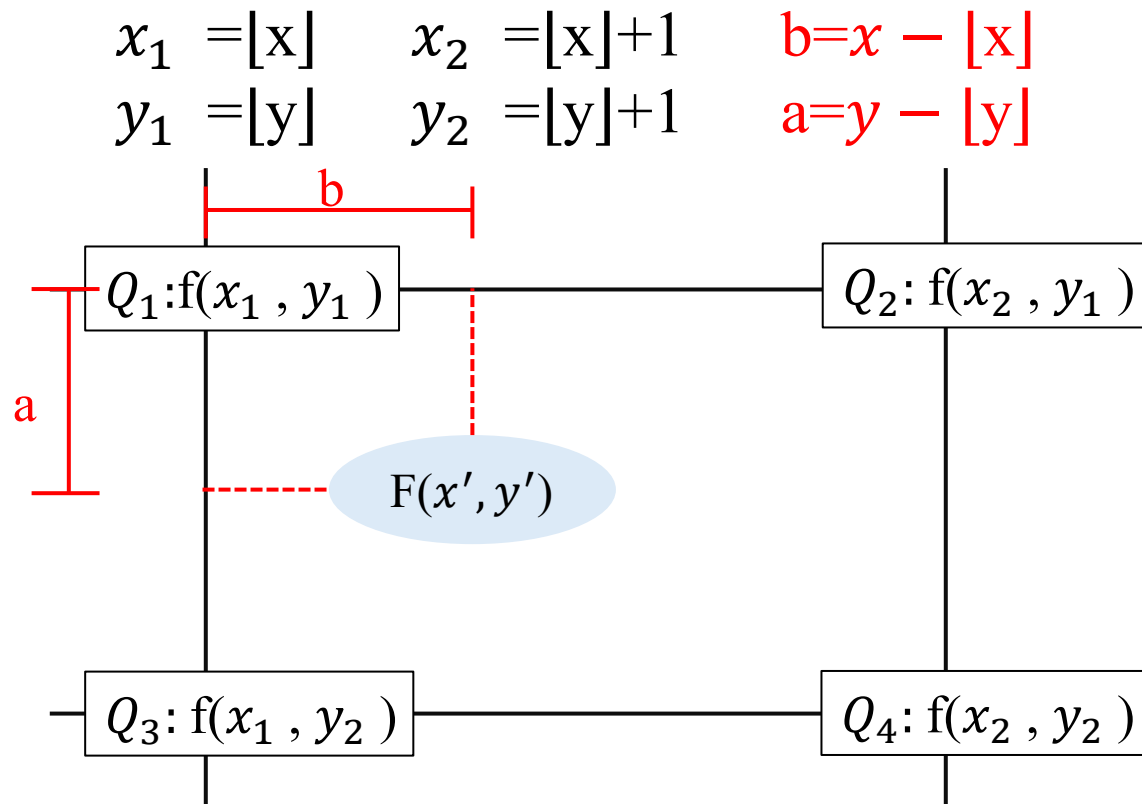
Step2:映射到原圖

要算目標圖 $F(x', y')$ 的值
先算映射到原圖的 $f(x, y)$

$$x = x' \times S_x$$

$$y = y' \times S_y$$

Step3:計算目標圖值



$$F(x', y') = (1-b) \times [(1-a) \times Q_1 + a \times Q_3] + b \times [(1-a) \times Q_2 + a \times Q_4]$$



2-D 雙線性內插法 範例Example 2

Question: Using 2-D bilinear interpolation, find the value of the pixel marked ‘?’ in the 4×4 image.

3×3

10	20	30
40	50	60
70	80	90



4×4

	?		

Step1:計算縮放比例

- 原始圖像尺寸:寬度 $w_1 = 3$, 高度 $h_1 = 3$
- 目標圖像尺寸:寬度 $w_2 = 4$, 高度 $h_2 = 4$

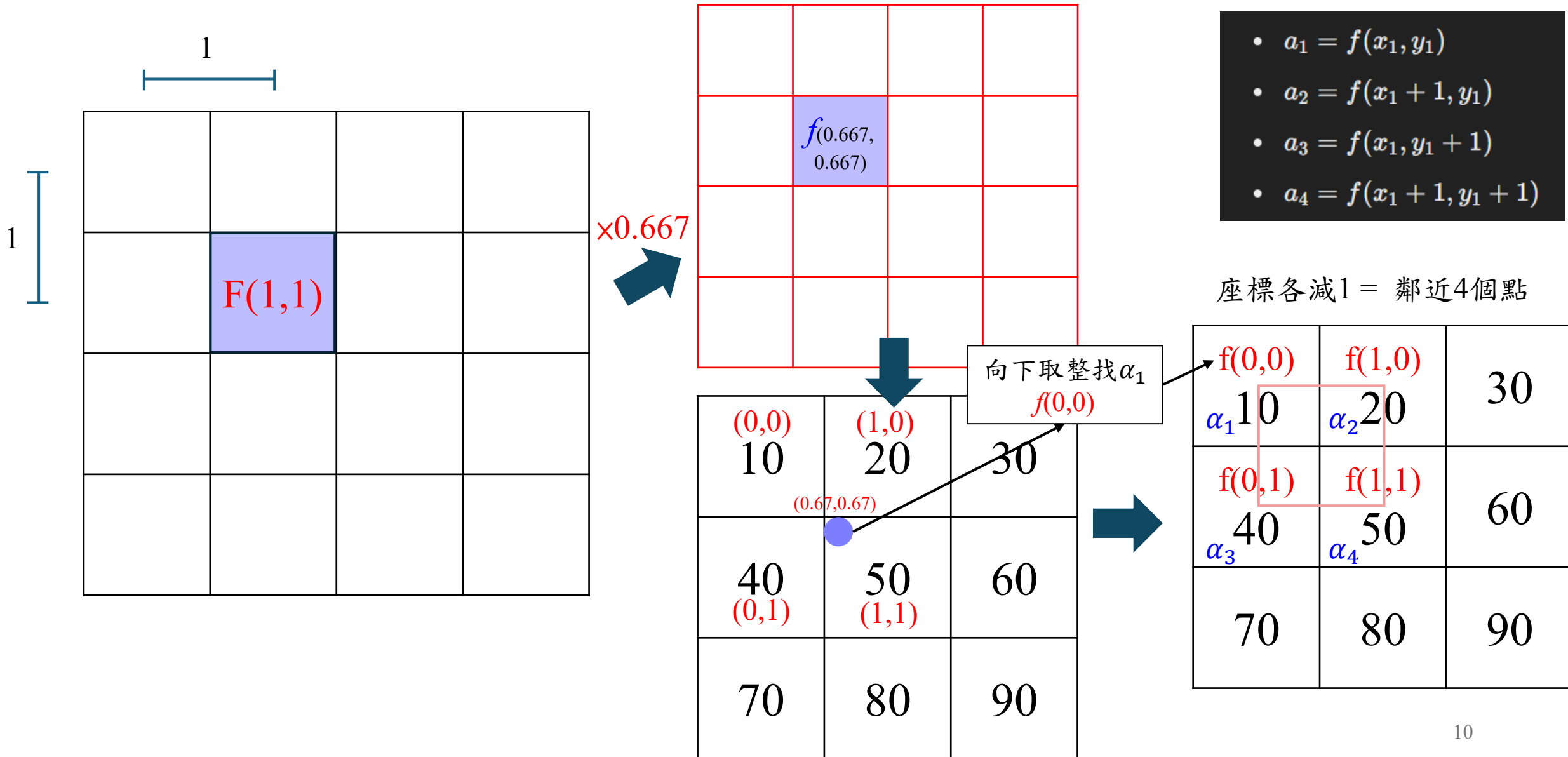
$$S_x = \frac{w_1 - 1}{w_2 - 1} = \frac{3 - 1}{4 - 1} = \frac{2}{3} \approx 0.667$$

$$S_y = \frac{h_1 - 1}{h_2 - 1} = \frac{2}{3} \approx 0.667$$

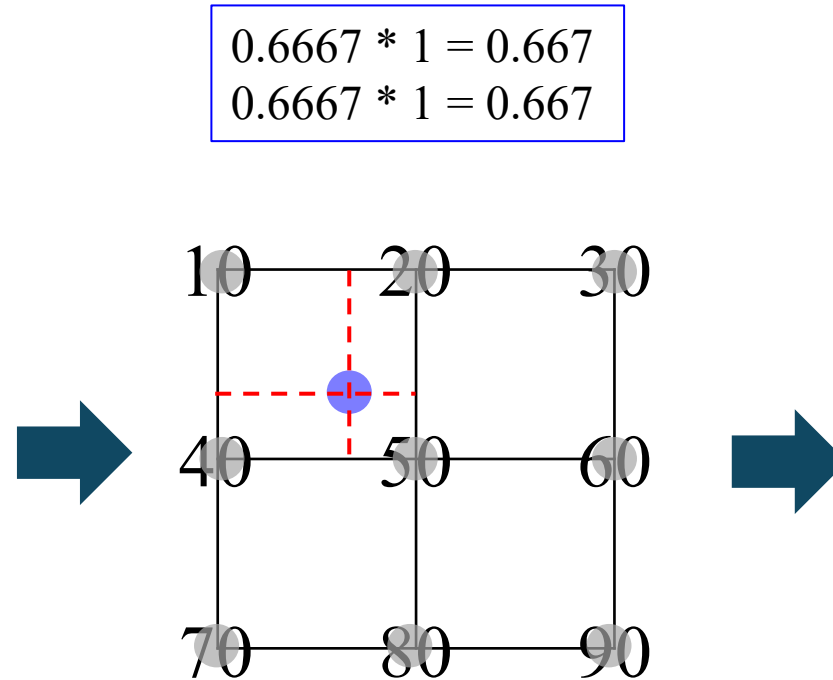
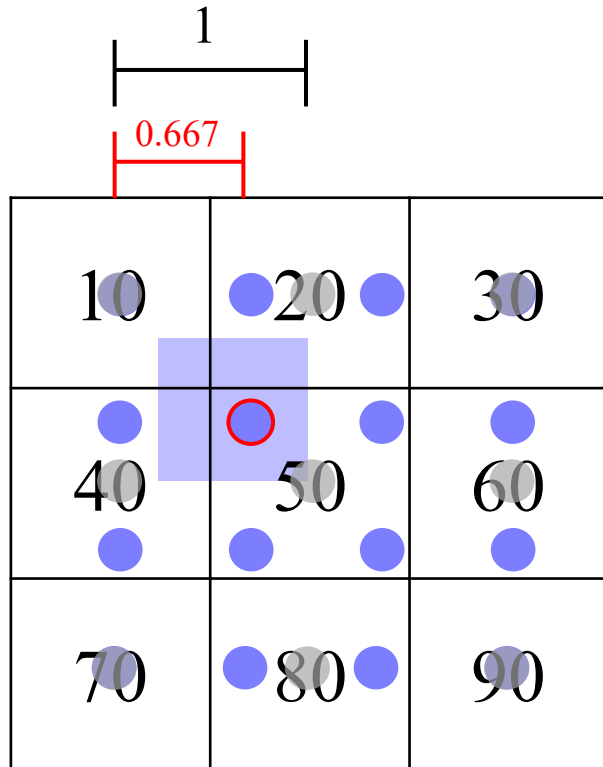
這個就是縮放後
Pixel與Pixel間的距離

F: 目標圖
f: 原圖

Step2:找鄰近四個點 $\alpha_1 \sim \alpha_4$

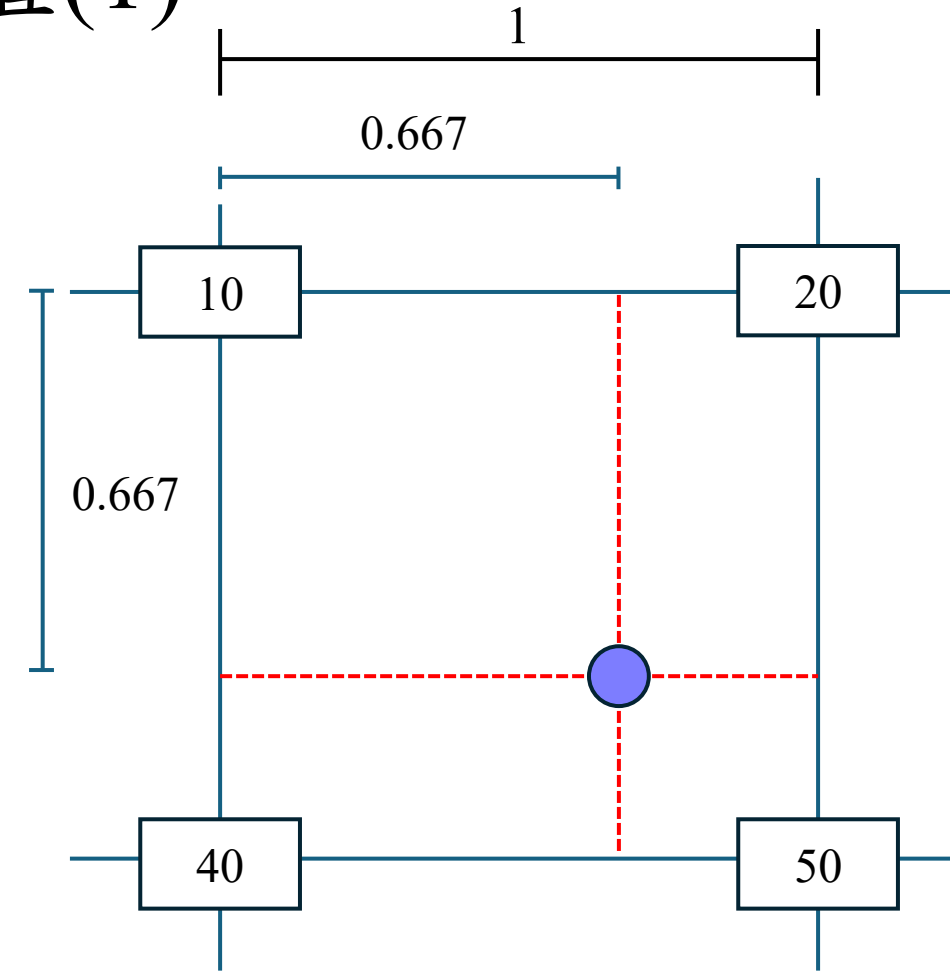


Step3:計算內插值(1)



$$0.6667 * 1 = 0.667$$

$$0.6667 * 1 = 0.667$$



$$F(1,1) = (1-0.6667) \times [(1-0.6667) \times 10 + 0.6667 \times 40] + 0.6667 \times [(1-0.6667) \times 20 + 0.6667 \times 50] = 36.6672$$



練習

Question: Using 2-D bilinear interpolation, find the value of the pixel marked '?' in the 4×4 image.

3×3

10	20	30
40	50	60
70	80	90



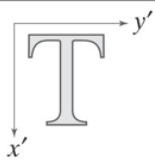
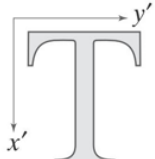
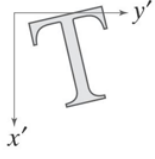
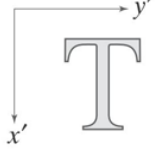
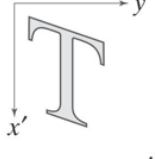
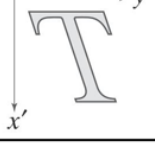
4×4

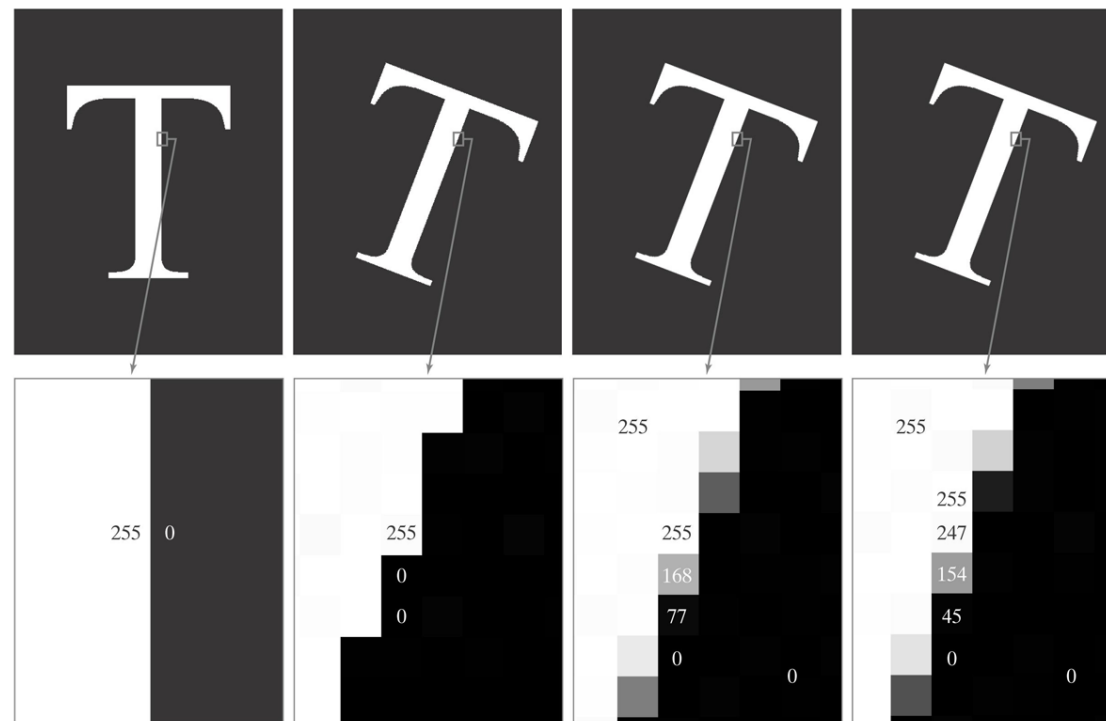
		?	

Ans: 63.36

2.5 空間運算：仿射轉換

表 2.1 依據 (2-23) 式的仿射轉換

轉換名稱	仿射矩陣 A	座標方程式	例 子
恆等式	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
尺度調整 / 翻轉 (對於翻轉，將一個尺度因子設成 -1 且其它設成 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
旋轉 (對原點)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos\theta - y \sin\theta$ $y' = x \sin\theta + y \cos\theta$	
平移	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
切變 (垂直)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
切變 (水平)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	



a b c d
e f g h

圖 2.29 (a) 字母 T 的 541×421 影像；(b) 用最近鄰插補法做強度指定給經旋轉 -21° 的影像；(c) 用雙線性插補法使影像旋轉 -21° ；(d) 用雙立方插補法使影像旋轉 -21° 。(e)~(h) 放大部分 (每個方形是一個像素，而所顯示的數字是強度值)。

範例一:仿射轉換

- Q: ➤ 寫出 **平移**、**縮放**、**旋轉** 的仿射矩陣
- 寫出 **反平移**、**反縮放**、**反旋轉** 的仿射矩陣

(可能抽幾個考而已)
(pick a few to test)

- Write the affine matrices: **translation**, **scaling**, **rotation**
- Write the **inverse affine** matrices: translation, scaling, rotation

Ans:

平移
translation

$$T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

反平移
inverse
translation

$$T^{-1} = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix}$$

縮放
scaling

$$S = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

反縮放
inverse
scaling

$$S^{-1} = \begin{bmatrix} \frac{1}{s_x} & 0 & 0 \\ 0 & \frac{1}{s_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

旋轉
rotation

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

反旋轉
inverse
rotation

$$R(\theta)^{-1} = R(-\theta) = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

範例二:正逆仿射轉換

- 影像 **先** 縮放 ($S_x=3, S_y=5$) , **再** 旋轉 $\theta = -90^\circ$ 。
- 請寫出其 仿射矩陣 與 逆仿射矩陣 。
- The image is first **scaled** ($S_x=3, S_y=5$), then **rotated** -90° .
- Write its affine matrix and inverse affine matrix .

正向仿射矩陣
(affine matrix)

再轉R 先縮放S ANS:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R(-90^\circ)} \underbrace{\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{S(3,5)} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & 5 & 0 \\ -3 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R(\theta)} \underbrace{\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{S(S_x, S_y)}$$

逆向仿射矩陣
(Inverse affine matrix)

再反縮放 先反轉 ANS:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{S^{-1}} \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R(+90^\circ)} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \boxed{\begin{bmatrix} 0 & -\frac{1}{3} & 0 \\ \frac{1}{5} & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} \frac{1}{S_x} & 0 & 0 \\ 0 & \frac{1}{S_y} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{S^{-1}} \underbrace{\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R(-\theta)}$$

第3章:直方圖等化、相關性、卷積

Chapter 3: Histogram Equalization, Correlation, and Convolution

符號定義

- k = 第幾個灰階 (which gray level)
- r_k : 原始灰階值 (original gray level)
- n_k : 該灰階出現的次數 (number of pixels with gray level r_k)
- N : 總像素數 (total number of pixels)
- $PMF(r_k)$: 機率質量函數 $= \frac{n_k}{N}$ (probability of gray level r_k)
- $CDF(r_k)$: 累積分布函數 $= \sum_{i=1}^k PMF(r_i)$ (cumulative probability)
- $s_{\max}, s_{\min}$: 輸出灰階範圍的最大與最小值 (max & min output gray levels)
- s_k : 等化後的新灰階值 (new gray level after equalization)
- $s_k = round(CDF(r_k) \times (s_{\max} - s_{\min}) + s_{\min})$

EX1:直方圖等化(Histogram Equalization)

- 一張4階的 $3 \times 3 = 9$ 個像素的灰階影像
- 運用直方圖等化(HE)，找出轉換後的像素值 $s = T(r)$ 。
- A 4-level grayscale image with $3 \times 3 = 9$ pixels is given.
- Find the equalized values $s = T(r)$.



➤ $r_1 = 0$, $r_2 = 1$, $r_3 = 2$, $r_4 = 3$

➤ $n_1 = 4$, $n_2 = 2$, $n_3 = 2$, $n_4 = 1$



r_1 : 總共有4個
(number of pixels = 4)

r_4	r_1	r_3	3	0	2
r_1	r_2	r_3	0	1	2
r_1	r_2	r_1	0	1	0

EX1:直方圖等化(HE) (1/3) 算PMF

- 一張4階的 $3 \times 3 = 9$ 個像素的灰階影像
- 運用直方圖等化(HE)，找出轉換後的像素值 $s = T(r)$ 。
- A 4-level grayscale image with $3 \times 3 = 9$ pixels is given.
- Find the equalized values $s = T(r)$.

$$PMF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$PMF(r_2) = \frac{n_2}{N} = \frac{2}{9}$$

$$PMF(r_3) = \frac{n_3}{N} = \frac{2}{9}$$

$$PMF(r_4) = \frac{n_4}{N} = \frac{1}{9}$$

➤ $n_1 = 4, n_2 = 2, n_3 = 2, n_4 = 1$

會使用到的公式：

$$PMF(r_k) = \frac{n_k}{N}$$

EX1:直方圖等化(HE) (2/3) 算CDF

$$CDF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$CDF(r_2) = \frac{n_1}{N} + \frac{n_2}{N} = \frac{6}{9}$$

$$CDF(r_3) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} = \frac{8}{9}$$

$$CDF(r_4) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} + \frac{n_4}{N} = \frac{9}{9}$$

$$PMF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$PMF(r_2) = \frac{n_2}{N} = \frac{2}{9}$$

$$PMF(r_3) = \frac{n_3}{N} = \frac{2}{9}$$

$$PMF(r_4) = \frac{n_4}{N} = \frac{1}{9}$$

會使用到的公式：

$$CDF(r_k) = \sum_{i=1}^k PMF(r_i)$$

EX1:直方圖等化(HE) (3/3) 算 $s=T(r)$

$$s_k = \text{round}(CDF(r_k) \times (s_{\max} - s_{\min}) + s_{\min})$$

$$s_1 = \text{round}(CDF(r_1) \times (3 - 0)) = \text{round}\left(\frac{4}{9} \times 3\right) = \text{round}(1.33) \approx 1$$

$$s_2 = \text{round}(CDF(r_2) \times (3 - 0)) = \text{round}\left(\frac{6}{9} \times 3\right) = \text{round}(2) \approx 2$$

$$s_3 = \text{round}(CDF(r_3) \times (3 - 0)) = \text{round}\left(\frac{8}{9} \times 3\right) = \text{round}(2.66) \approx 3$$

$$s_4 = \text{round}(CDF(r_4) \times (3 - 0)) = \text{round}\left(\frac{9}{9} \times 3\right) = \text{round}(3) \approx 3$$

$$CDF(r_1) = \frac{n_1}{N} = \frac{4}{9}$$

$$CDF(r_2) = \frac{n_1}{N} + \frac{n_2}{N} = \frac{6}{9}$$

$$CDF(r_3) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} = \frac{8}{9}$$

$$CDF(r_4) = \frac{n_1}{N} + \frac{n_2}{N} + \frac{n_3}{N} + \frac{n_4}{N} = \frac{9}{9}$$

	$k=1$	$k=2$	$k=3$	$k=4$
原始強度 r_k	0	1	2	3
轉換後強度 s_k	1	2	3	3

會使用到的公式：

$$s_k = \text{round}(CDF(r_k) \times (s_{\max} - s_{\min}) + s_{\min})$$

#



Extra Question

- 把算好的等化後灰階值 s_k 填回原始影像。
- Fill the equalized gray values s_k back into the original image.

	$k=1$	$k=2$	$k=3$	$k=4$
原始強度 r_k	0	1	2	3
轉換後強度 s_k	1	2	3	3

Before HE

3	0	2
0	1	2
0	1	0



After HE

3	1	3
1	2	3
1	2	1

#

Example1:相關性運算

5×5 大小影像

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

*

3×3 kernel

1	0	1
0	1	0
1	0	1

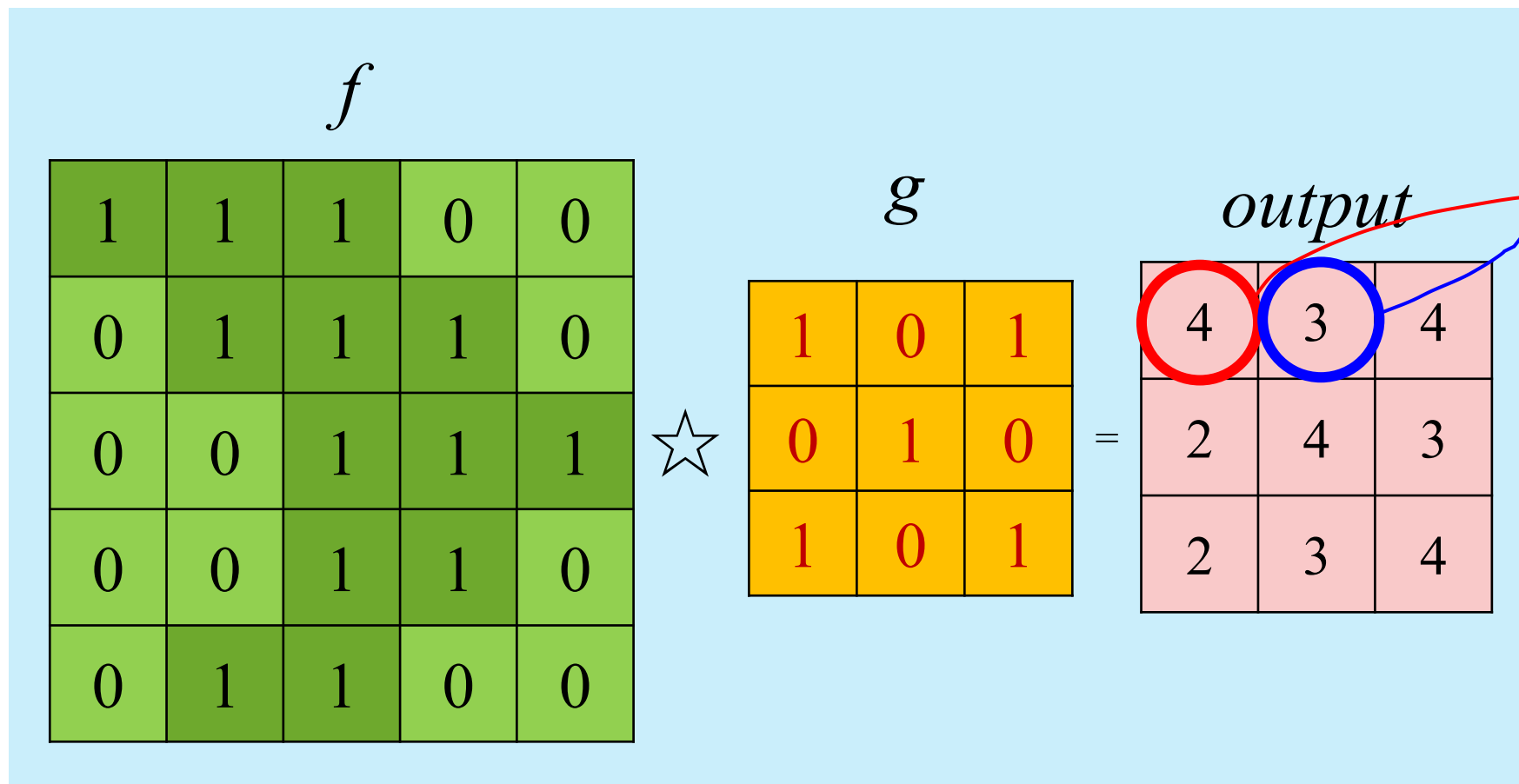
=

output

?	?	?
?	?	?
?	?	?

相關性運算

- 有一個影像 f 為大小為 $5*5$ ，遮罩 g 大小為 $3*3$ ，求算 f 對 g 相關性運算，即 $f \star g$ ，padding = 0



1	1	0
1	1	1
0	1	1

1	0	1
0	1	0
1	0	1

做點積，同位置相乘，
之後全部相加

1	1	1
0	1	1
0	0	1

1	0	1
0	1	0
1	0	1

Dot product: multiply matching
elements, then sum the results.

補空白Pad

影像

7	8	9
12	13	14
17	18	19

Padding=1 影像
(P=2時 會補兩圈0)

0	0	0	0	0
0	7	8	9	0
0	12	13	14	0
0	17	18	19	0
0	0	0	0	0

Kernel

1	0	0
0	1	0
0	0	1

Padding後可以
往更左上角開始算
較不會損失邊緣特徵

0	0	0	0
0	7	8	9
0	12	13	14
0	17	18	19

卷積(Convolution)

- 有一個影像 f 為大小為 $3*3$ ，遮罩 g 大小為 $3*3$ ，求算 f 對 g 卷積，即 $f*g$ ，padding = 1

$$\begin{array}{|c|c|c|} \hline f & & \\ \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline \end{array} * \begin{array}{|c|c|c|} \hline g & & \\ \hline 1 & 2 & 3 \\ \hline 4 & 5 & 6 \\ \hline 7 & 8 & 9 \\ \hline \end{array}$$

g 上下顛倒，左右翻左後 成為 g'

f				
0	0	0	0	0
0	1	2	3	0
0	4	5	6	0
0	7	8	9	0
0	0	0	0	0

*

g'		
9	8	7
6	5	4
3	2	1

=

$output$		
26	56	54
84	165	144
134	236	186

0	0	0	9	8	7
0	1	2	6	5	4
0	4	5	3	2	1

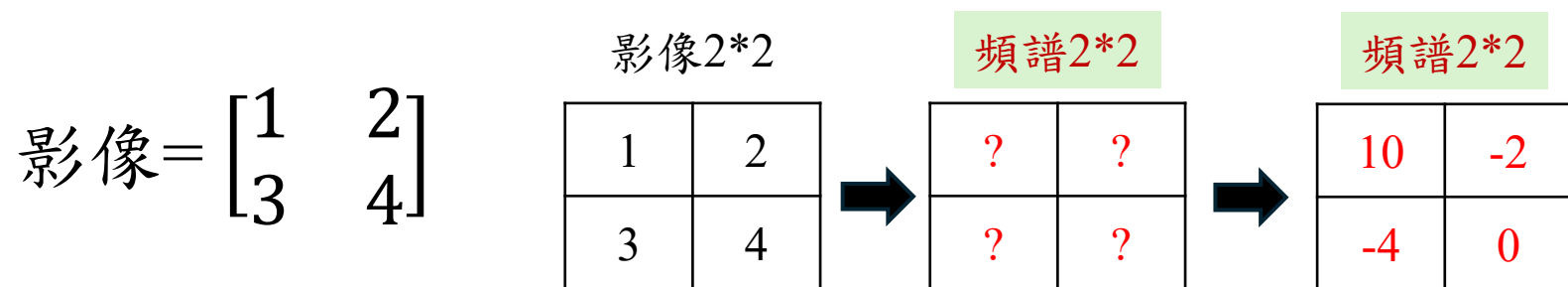
Dot product: multiply matching elements, then sum the results.

第4章:傅立葉變換

Chapter 4: Fourier Transform

Example 1: 傅立葉變換

- 要將一張寬×高=2×2的影像進行傅立葉轉換，我們可以使用離散傅立葉變換（DFT）公式來計算其頻譜。求傅立葉變換後的 $F(0,0)$



會使用到的公式：

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

座標：

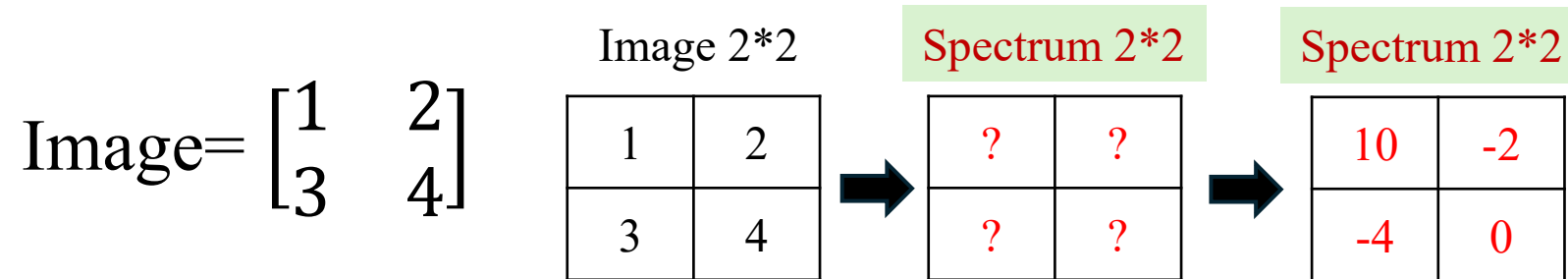
(0,0)	(1,0)
(0,1)	(1,1)

$$F(0,0) = f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 1}{2}\right)}$$

$$F(0,0) = 1 \cdot 1 + 3 \cdot 1 + 2 \cdot 1 + 4 \cdot 1 = 10$$

Example 1: Fourier Transform

- To perform a Fourier Transform on a 2x2 image, we can use the Discrete Fourier Transform (DFT) formula to compute its spectrum. Find the value of $F(0,0)$ after the Fourier Transform.



The formula to be used:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Coordinate:

(0,0)	(1,0)
(0,1)	(1,1)

$$F(0,0) = f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{2} + \frac{0 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{2} + \frac{0 \cdot 1}{2}\right)}$$

$$F(0,0) = 1 \cdot 1 + 3 \cdot 1 + 2 \cdot 1 + 4 \cdot 1 = 10$$

Example 2: Fourier Transform

頻譜2*2

1	2
3	4



Spectrum 2*2

	F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

Image 2*2	1	2	Spectrum 2*2	10	
	3	4			F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(1,1) = f(0,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)}$$

$$e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} = e^{-j2\pi\left(\frac{2}{2}\right)} = e^{-j2\pi} \quad \text{Euler's formula!} \quad e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$(\theta = -2\pi)$$

$$e^{-j2\pi} = \cos(-2\pi) + j \sin(-2\pi) = 1 + j0 = 1$$

$$e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} = e^{-j2\pi\left(\frac{1}{2}\right)} = e^{-j\pi}$$

$$(\theta = -\pi)$$

$$e^{-j\pi} = \cos(-\pi) + j \sin(-\pi) = -1 + j0 = -1$$

Image 2*2	1	2
	3	4

Spectrum 2*2	10	
		F(1,1)?

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

$$F(1,1) = f(0,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 0}{2}\right)} + f(0,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 0}{2} + \frac{1 \cdot 1}{2}\right)} + f(1,0) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} + f(1,1) \cdot e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)}$$

$$e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 0}{2}\right)} = e^{-j2\pi\left(\frac{1}{2}\right)} = e^{-j\pi}$$

Euler's formula!

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$(\theta = -\pi)$$

$$e^{-j\pi} = \cos(-\pi) + j \sin(-\pi) = -1 + j0 = -1$$

$$e^{-j2\pi\left(\frac{1 \cdot 1}{2} + \frac{1 \cdot 1}{2}\right)} = e^{-j2\pi\left(\frac{2}{2}\right)} = e^{-j2\pi} = \cos(-2\pi) + j \sin(-2\pi) = 1 + j0 = 1$$

$$F(1,1) = 1 \cdot (1) + 3 \cdot (-1) + 2 \cdot (-1) + 4 \cdot (1) = 0$$

頻譜2*2

10	?
?	0

Example 3 : Fourier Transform

Image $f(x,y)$ 3*3

1	2	3
4	5	6
7	8	9

Spectrum $|F(u,v)|$ 3*3

$ F(0,1) $		

Q: Find the magnitude of $|F(0,1)|$?

Fourier Transform

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Magnitude Spectrum

$$Magnitude = |F(u,v)| = \sqrt{\text{實部}(F(u,v))^2 + \text{虛部}(F(u,v))^2}$$

$$|F(u,v)| = \sqrt{(Re(F(u,v)))^2 + (Im(F(u,v)))^2}$$

Fourier Spectrum

Fourier Transform

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

1 radian = $180/\pi$ degrees

$$\begin{aligned}
 F(0,1) &= f(0,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 0}{3}\right)} & + f(1,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 0}{3}\right)} & + f(2,0) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 0}{3}\right)} \\
 &+ f(0,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 1}{3}\right)} & + f(1,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 1}{3}\right)} & + f(2,1) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 1}{3}\right)} \\
 &+ f(0,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 0}{3} + \frac{1 \cdot 2}{3}\right)} & + f(1,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 1}{3} + \frac{1 \cdot 2}{3}\right)} & + f(2,2) \cdot e^{-j2\pi\left(\frac{0 \cdot 2}{3} + \frac{1 \cdot 2}{3}\right)}
 \end{aligned}$$



$$\begin{aligned}
 F(0,1) &= 1 \cdot e^{-j\left(\frac{0\pi}{3}\right)} & + 2 \cdot e^{-j\left(\frac{0\pi}{3}\right)} & + 3 \cdot e^{-j\left(\frac{0\pi}{3}\right)} \\
 &+ 4 \cdot e^{-j\left(\frac{2\pi}{3}\right)} & + 5 \cdot e^{-j\left(\frac{2\pi}{3}\right)} & + 6 \cdot e^{-j\left(\frac{2\pi}{3}\right)} \\
 &+ 7 \cdot e^{-j\left(\frac{4\pi}{3}\right)} & + 8 \cdot e^{-j\left(\frac{4\pi}{3}\right)} & + 9 \cdot e^{-j\left(\frac{4\pi}{3}\right)}
 \end{aligned}$$

$$\frac{4 \cdot 180}{3} = 240^\circ$$

Fourier Spectrum

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$\begin{aligned} F(0,1) = & 1 * (\cos(-0^\circ) + j\sin(-0^\circ)) & +2 * (\cos(-0^\circ) + j\sin(-0^\circ)) & +3 * (\cos(-0^\circ) + j\sin(-0^\circ)) \\ & + 4 * (\cos(-120^\circ) + j\sin(-120^\circ)) & +5 * (\cos(-120^\circ) + j\sin(-120^\circ)) & +6 * (\cos(-120^\circ) + j\sin(-120^\circ)) \\ & + 7 * (\cos(-240^\circ) + j\sin(-240^\circ)) & +8 * (\cos(-240^\circ) + j\sin(-240^\circ)) & +9 * (\cos(-120^\circ) + j\sin(-240^\circ)) \end{aligned}$$



$$\begin{aligned} F(0,1) = & 1 * (1 + j0) & +2 * (1 + j0) & +3 * (1 + j0) \\ & + 4 * (-0.5 + j(-0.866)) & +5 * (-0.5 + j(-0.866)) & +6 * (-0.5 + j(-0.866)) \\ & + 7 * (-0.5 + j(0.866)) & +8 * (-0.5 + j(0.866)) & +9 * (-0.5 + j(0.866)) \end{aligned}$$

Fourier Spectrum

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$F(0,1) = (1 + 0j)$$

$$+(-2 - j3.464)$$

$$+(-3.5 + j(6.062))$$

$$+(2 + j0)$$

$$+(-2.5 - j4.33)$$

$$+(-4 + j6.928)$$

$$+(3 + j0)$$

$$+(-3 - j5.196)$$

$$+(-4.5 + j7.794)$$

$$F(0,1) = -13.5 + 7.794j$$

$$\text{Magnitude} = |F(u, v)| = \sqrt{\text{實部}(F(u, v))^2 + \text{虛部}(F(u, v))^2}$$

$$|F(0,1)| = \sqrt{(-13.5)^2 + (7.794)^2} = 15.588 \#$$

Spectrum $|F(u, v)|$

45	5.196	5.196
15.588	0	0
15.588	0	0

#

重點公式整理

(Key formula summary)

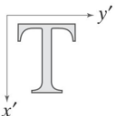
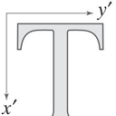
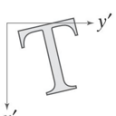
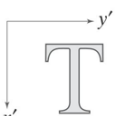
公式整理

索引

索引值 $S = x + My + MNz = x + M(y + Nz)$

- $z = S/MN$ (取商數)
- $y = S \bmod MN / M$ (取商數)
- $x = S \bmod MN \bmod M$

仿射

轉換名稱	仿射矩陣 A	座標方程式	例子
恆等式	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
尺度調整 / 翻轉 (對於翻轉，將一個尺度因子設成 -1 且其它設成 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
旋轉 (對原點)	$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos\theta - y \sin\theta$ $y' = x \sin\theta + y \cos\theta$	
平移	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	

值方圖等化

$$PMF(r_k) = \frac{n_k}{N}$$

$$CDF(r_k) = \sum_{i=1}^k PMF(r_i)$$

$$s_k = \text{round}(CDF(r_k) \times (s_{\max} - s_{\min}) + s_{\min})$$

相關性

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x+s, y+t)$$

卷積

$$(w \star f)(x, y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s, t) f(x-s, y-t)$$

傅立葉變換、頻譜

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) \cdot e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

$$e^{j\theta} = \cos(\theta) + j \sin(\theta)$$

$$\text{Magnitude} = |F(u, v)| = \sqrt{\text{實部}(F(u, v))^2 + \text{虛部}(F(u, v))^2}$$

公式整理 2-D 雙線性內插法

Step1:縮放比例 (Scaling Ratio)

- 原始圖像尺寸:寬度 w_1 , 高度 h_1
- 目標圖像尺寸:寬度 w_2 , 高度 h_2

$$S_x = \frac{w_1 - 1}{w_2 - 1} \quad S_y = \frac{h_1 - 1}{h_2 - 1}$$

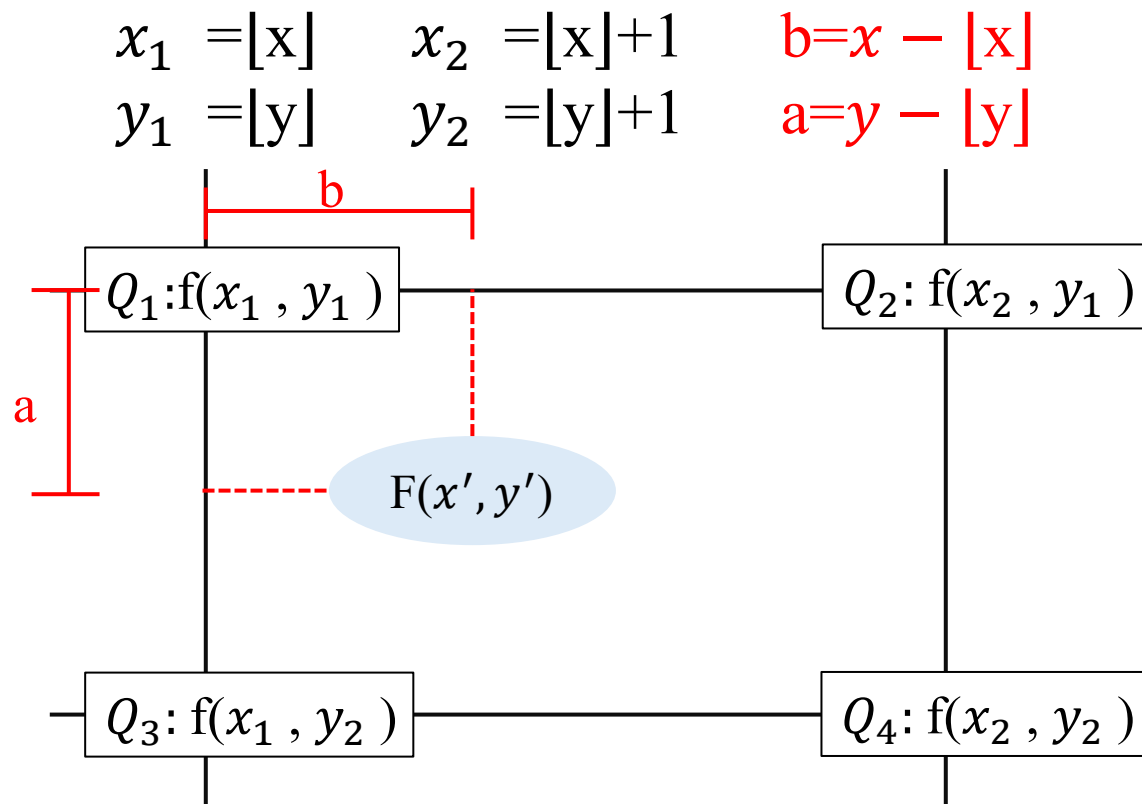
Step2:映射到原圖

要算目標圖 $F(x', y')$ 的值
先算映射到原圖的 $f(x, y)$

$$x = x' \times S_x$$

$$y = y' \times S_y$$

Step3:計算目標圖值



$$F(x', y') = (1-b) \times [(1-a) \times Q_1 + a \times Q_3] + b \times [(1-a) \times Q_2 + a \times Q_4]$$